Arjuna JEE 2025

Maths

Sequence & Series

DPP: 1

Q1 If 100 times the 100^{th} terms of an $\mathrm{A.P.}$ with non-zero common difference equals the 50 times its 50^{th} term, then the 150^{th} term of this $\mathrm{A.P.}$ is

(A) 150
(B) zero
(C) -150
(D) 150 times its 50^{th} term

Q2 If the 19^{th} term of a non-zero $\mathrm{A.P.}$ is zero then the ratio $\left(49^{\text{th}}\text{ term}\right):\left(29^{\text{th}}\text{ term}\right)$ is

(A) $4:1$ (B) $1:3$
(C) $3:1$ (D) $2:1$

Q3 If $a_1, a_2, a_3 \ldots\ldots\ldots,$ are in Arithmetic Progression then $a_1 - 2a_2 + a_3 =$

(A) 2
(B) 1
(C) 0
(D) Cannot be determined

Q4 The first negative term of the sequence $2000, 1995, 1990, 1985, \ldots\ldots.$ is

(A) 401 (B) 402
(C) 403 (D) 404

Q5 The total number of integers between 100 and 1000 that is not divisible by 7 is

(A) 899 (B) 128
(C) 771 (D) 770

Q6 The number of common terms to the sequence $3, 5, 7, 9 \ldots\ldots\ldots,$ upto 50 terms, $2, 5, 8, 11, \ldots\ldots\ldots,$ upto 60 terms are

(A) 16 (B) 17
(C) 15 (D) 18

Q7 If $\log_5 2, \log_5(2^x+1)$ and $\log_5(2^x+5/2)$ are in $\mathrm{A.P.}$ then x is equal to

(A) 1 (B) 2
(C) 3 (D) 4

Q8 The sum of three numbers in $\mathrm{A.P.}$ is 27 and the sum of their square is 293. Then numbers are

(A) $5, 10, 15$
(B) $4, 9, 14$
(C) $3, 8, 13$
(D) $6, 11, 16$

Q9 The total number of integers between 1 and 200 which are multiple of 3 and 7 is

Q10 The number of identical terms in the sequence $2, 5, 8, 11, \ldots\ldots\ldots$ up to 100 terms and $3, 5, 7, 9, \ldots\ldots\ldots$ up to 100 terms, are ________

Q11 If $a_1, a_2, a_3 \ldots\ldots\ldots,$ are in $\mathrm{A.P.}$ where $a_2 = 76$ and $a_{12} = 26$ then least positive integer n for which $a_n < 0$ is ________

Q12 The 19^{th} term from end of progression $3, 6, 9, 12, \ldots\ldots, 72$ is ________.

Q13 If the sum of n terms of an $\mathrm{A.P.}$ is $2n + 3n^2$. Then its r^{th} term is given by

(A) $6r-1$ (B) $6r+1$
(C) $5r-1$ (D) $5r+1$

Q14 The sum of the integers between 1 and 200, which are multiple of 3 or 7 is

(A) 2847 (B) 6633

(C) 945 (D) 8530

Q15 If $\frac{3+5+7+\ldots..+\text{ upto } n \text{ terms}}{5+8+11+\ldots.. \text{ upto 10 terms}}=7$ then $n=$

(A) 35 (B) 36

(C) 37 (D) 40

Q16 For an AP, $T_{10}=\frac{1}{20}$ and $T_{20}=\frac{1}{10}$ then $S_{200}=$

(A) $\frac{361}{2}$

(B) $\frac{201}{2}$

(C) $\frac{101}{2}$

(D) 50

Q17 Let $a_1, a_2, a_3 \ldots$ be term of AP If $\frac{a_1+a_2+\ldots..+a_p}{a_1+a_2+\ldots..+a_q}=\frac{P^2}{q^2}, p \neq q$ then $\frac{a_6}{a_{21}}=$

(A) $\frac{41}{11}$

(B) $\frac{7}{2}$

(C) $2/7$

(D) $\frac{11}{41}$

Q18 If the sum of 12^{th} and 22^{nd} terms of an AP is 100, then the sum of the first 33 terms of an AP is

(A) 1700 (B) 1650

(C) 3300 (D) 3400

Q19 If the first, second and last terms of an arithmetic series are a, b and c respectively, then the number of terms is

(A) $\frac{b+c-2a}{b-a}$

(B) $\frac{b+c+2a}{b-a}$

(C) $\frac{b+c-2a}{b+a}$

(D) $\frac{b+c+2a}{b+a}$

Answer Key

Q1	(B)
Q2	(C)
Q3	(C)
Q4	(B)
Q5	(C)
Q6	(B)
Q7	(A)
Q8	(B)
Q9	(A)
Q10	33
Q11	18
Q12	18
Q13	(A)
Q14	(D)
Q15	(A)
Q16	(B)
Q17	(D)
Q18	(B)
Q19	(A)